

THREADS

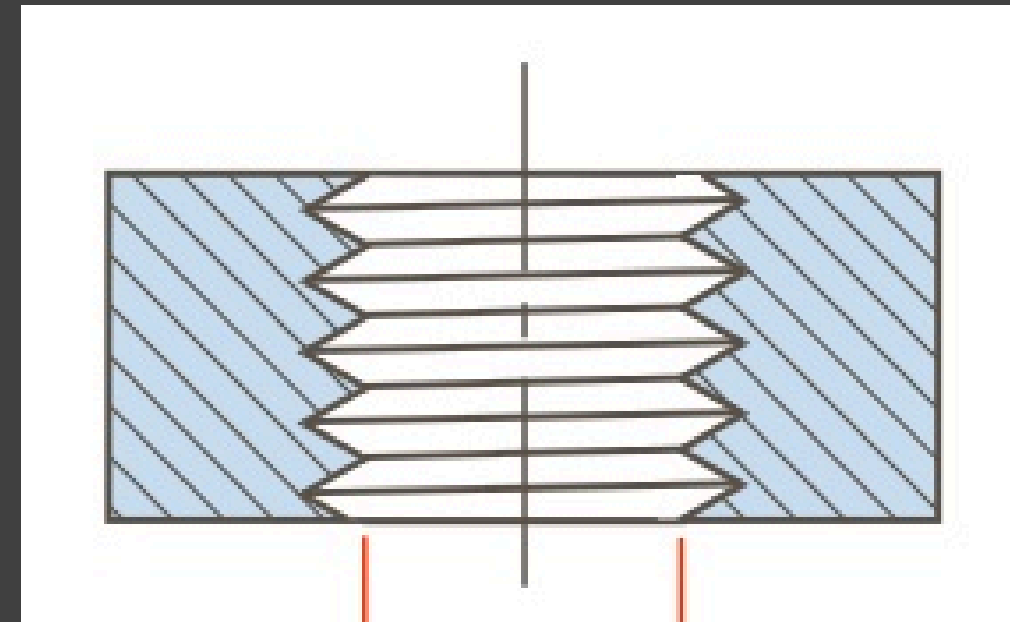
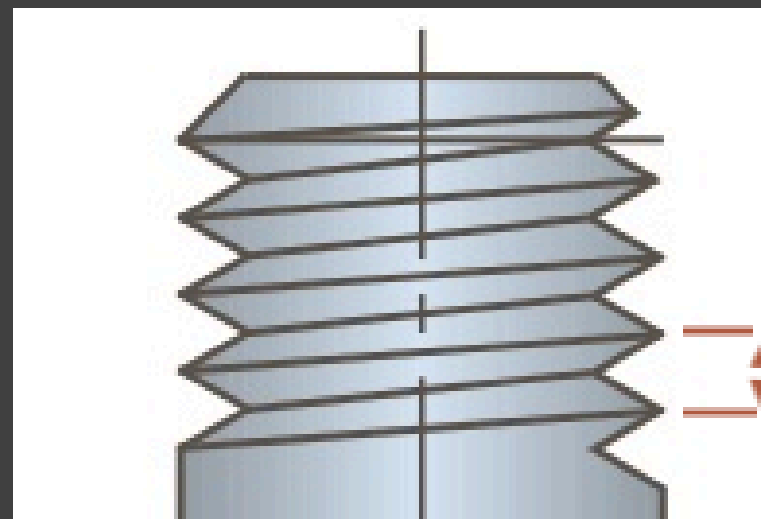
BY HARSHITA M.



WHAT ARE THREADS?

- Threads are spiral grooves that transform rotational motion into linear motion, or vice versa. They're crucial for fastening and movement in mechanical systems
- 2 different types
 - External Threads: Found on bolts or screws, like spirals on a shaft
 - Internal Threads: Found inside nuts or cylindrical components, like grooves in a hole

EXTERNAL THREADS



INTERNAL THREADS

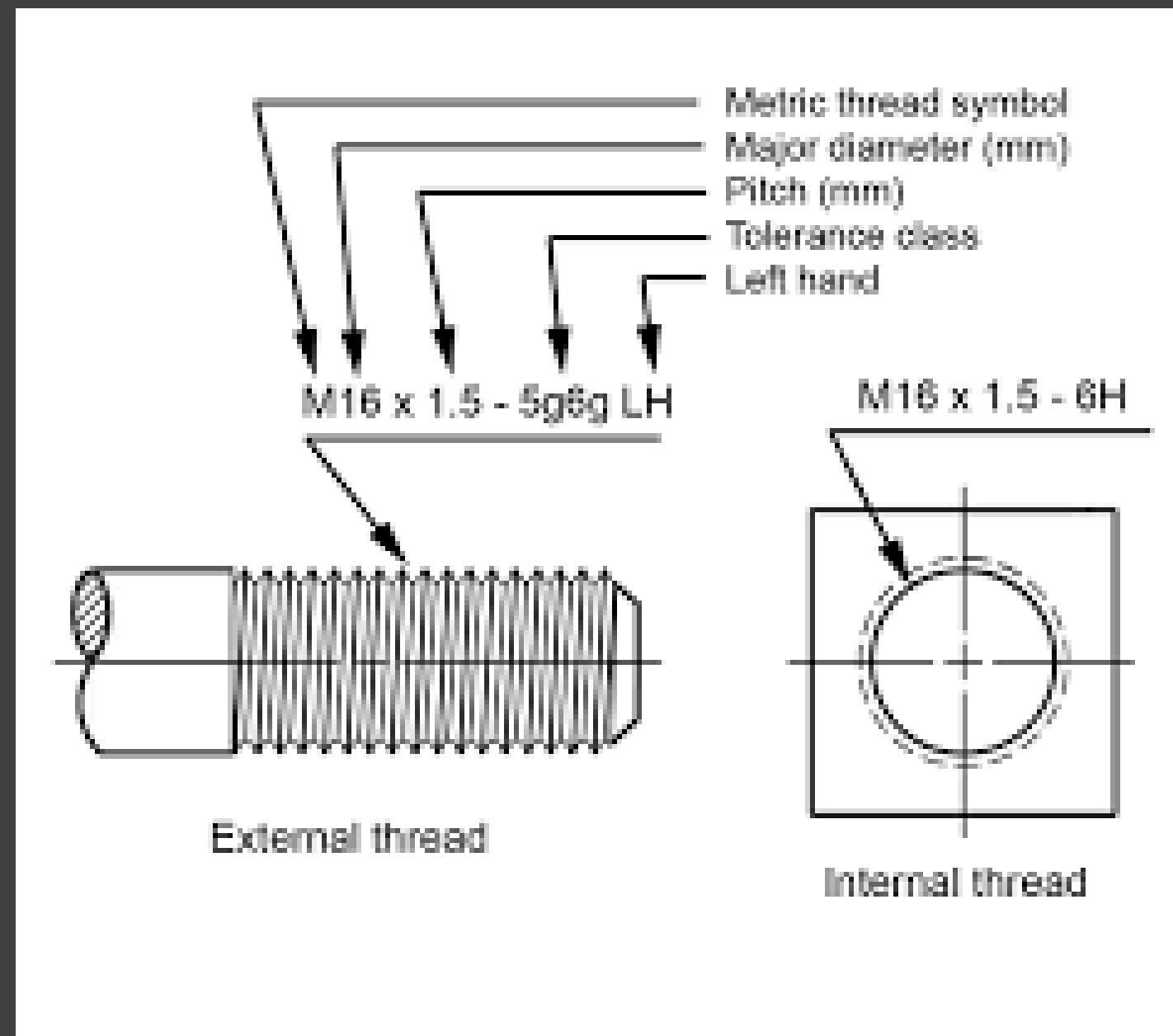
WHY ARE THEY IMPORTANT?

- **Threads aren't just for holding parts together—they enable precision, strength, and motion. Without them, mechanical assemblies like engines or robots would fail.**
- **Key functions**
 - **Securely join components.**
 - **Adjust and move parts with precision.**
 - **Distribute loads efficiently.**



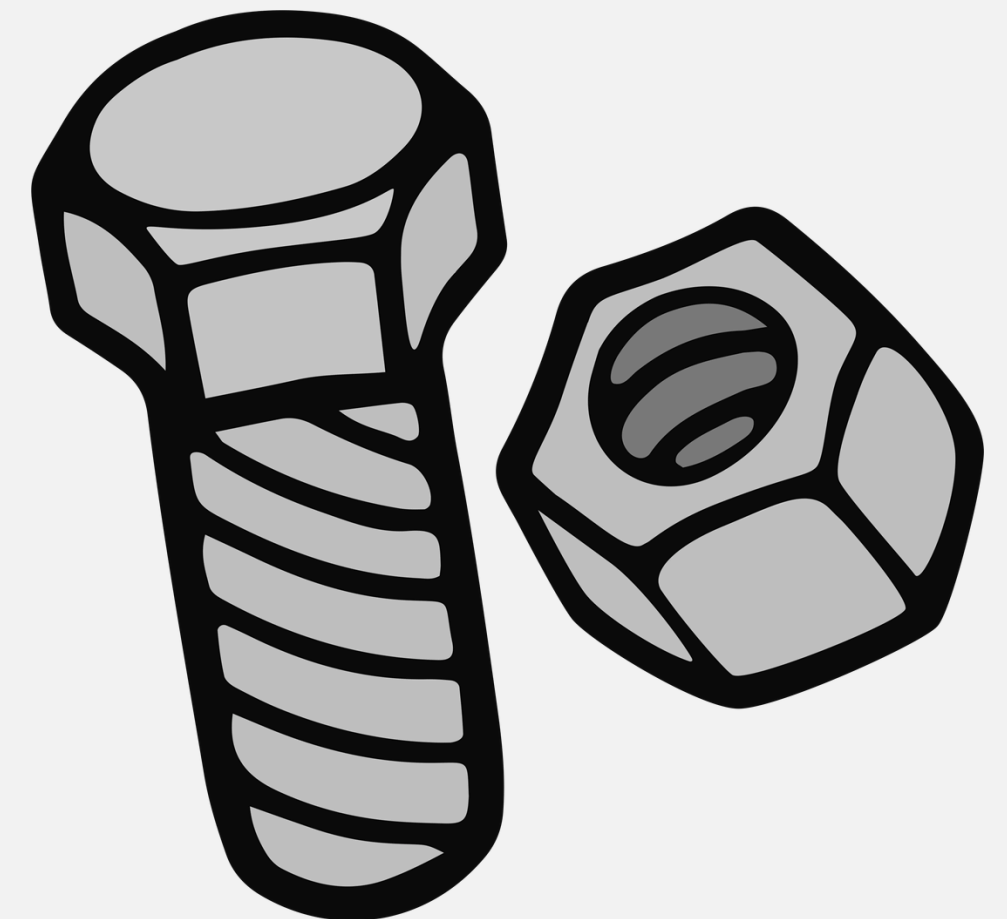
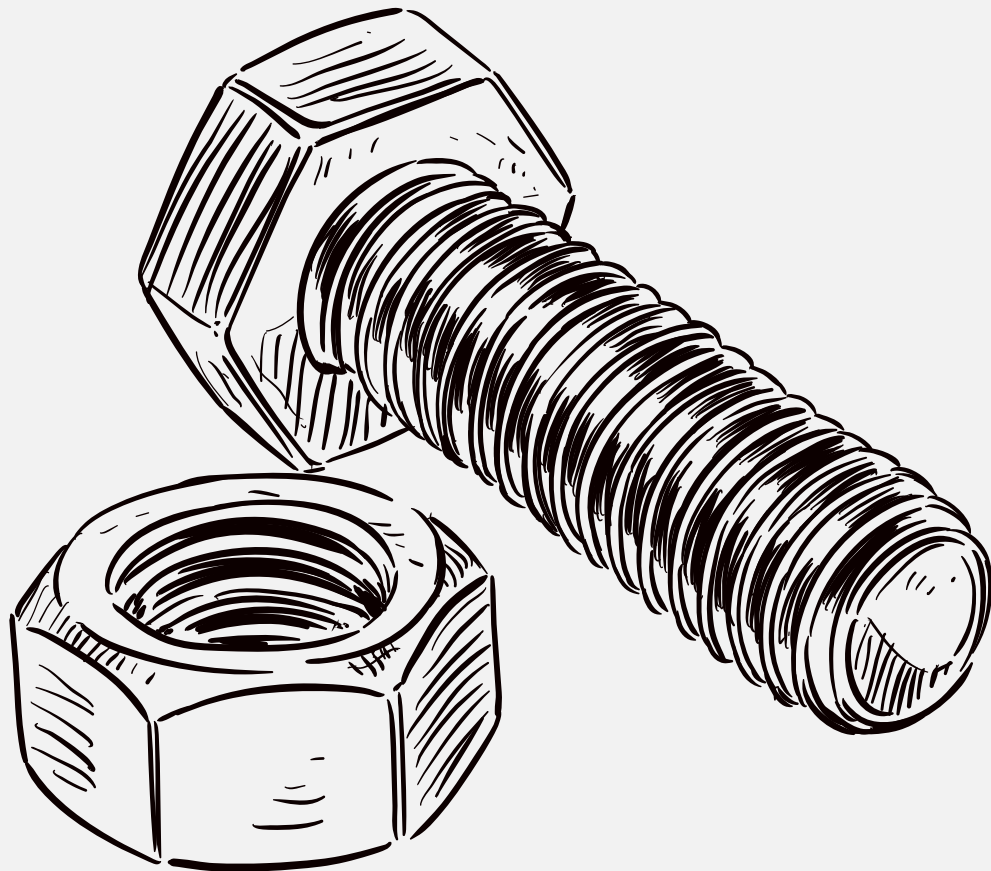
DECODING THREAD NOTES

- **Diameter:** The size of the thread (e.g., 10mm or ½ inch).
- **Pitch:** The distance between threads (e.g., 1.5mm or 20 threads per inch).
- **Direction:** Right-hand (standard) vs. left-hand threads.



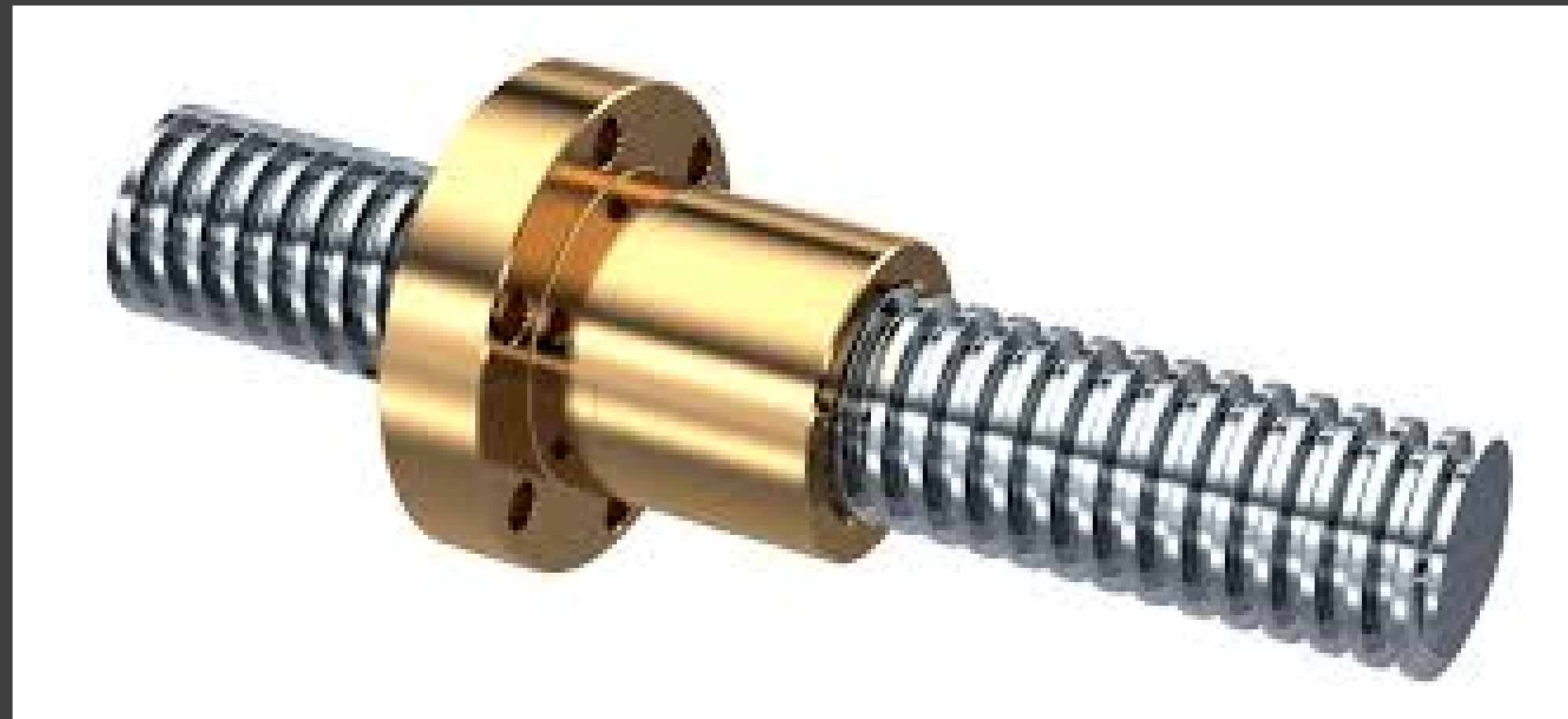
CHOOSING THE RIGHT FASTENER

- Picking the perfect threaded fastener isn't random—it's strategic.
- **Considerations:**
 - **Material:** Match the fastener to the strength and durability required.
 - **Thread Type:** Coarse for quick assembly, fine for high precision.
 - **Load and Environment:** Think about forces, vibration, and exposure to moisture or heat



THREADS IN ACTION

- Threads are everywhere—from everyday objects to high-tech prototypes.
- Applications:
 - Household: Screws in furniture, jars with threaded lids.
 - Engineering: Lead screws in CNC machines, bolts in bridges



SUMMARY

Threads are the backbone of mechanical fastening and precision movement. By understanding thread notes and how threads function, we can design more effective prototypes and products

